



Dacomitinib versus gefitinib as first-line treatment for patients with *EGFR*-mutation-positive non-small-cell lung cancer (ARCHER 1050): a randomised, open-label, phase 3 trial

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Summary

Background Dacomitinib is a second-generation, irreversible *EGFR* tyrosine kinase inhibitor. We compared its efficacy and safety with that of the reversible *EGFR* tyrosine kinase inhibitor gefitinib in the first-line treatment of patients with advanced *EGFR*-mutation-positive non-small-cell lung cancer (NSCLC).

Methods In this international, multicentre, randomised, open-label, phase 3 study (ARCHER 1050), we enrolled adults (aged ≥ 18 years or ≥ 20 years in Japan and South Korea) with newly diagnosed advanced NSCLC and one *EGFR* mutation (exon 19 deletion or Leu858Arg) at 71 academic medical centres and university hospitals in seven countries or special administrative regions. We randomly assigned participants (1:1) to receive oral dacomitinib 45 mg/day (in 28-day cycles) or oral gefitinib 250 mg/day (in 28-day cycles) until disease progression or another discontinuation criterion was met. Randomisation, stratified by race and *EGFR* mutation type, was done with a computer-generated random code assigned by a central interactive web response system. The primary endpoint was progression-free survival assessed by masked independent review in the intention-to-treat population. Safety was assessed in all patients who received at least one dose of study treatment. This study is registered with ClinicalTrials.gov, number NCT01774721, and is ongoing but no longer recruiting patients.

Findings Between May 9, 2013, and March 20, 2015, 452 eligible patients were randomly assigned to receive dacomitinib ($n=227$) or gefitinib ($n=225$). Median duration of follow-up for progression-free survival was 22.1 months (95% CI 20.3–23.9). Median progression-free survival according to masked independent review was 14.7 months (95% CI 11.1–16.6) in the dacomitinib group and 9.2 months (9.1–11.0) in the gefitinib group (hazard ratio 0.59, 95% CI 0.47–0.74; $p<0.0001$). The most common grade 3–4 adverse events were dermatitis acneiform (31 [14%] of 227 patients given dacomitinib vs none of 224 patients given gefitinib), diarrhoea (19 [8%] vs two [1%]), and raised alanine aminotransferase levels (two [1%] vs 19 [8%]). Treatment-related serious adverse events were reported in 21 (9%) patients given dacomitinib and in ten (4%) patients given gefitinib. Two treatment-related deaths occurred in the dacomitinib group (one related to untreated diarrhoea and one to untreated cholelithases/liver disease) and one in the gefitinib group (related to sigmoid colon diverticulitis/rupture complicated by pneumonia).

Interpretation Dacomitinib significantly improved progression-free survival over gefitinib in first-line treatment of patients with *EGFR*-mutation-positive NSCLC and should be considered as a new treatment option for this population.

Funding SFJ Pharmaceuticals Group and Pfizer.

Introduction

EGFR mutations are well known oncogenic driver mutations that occur in 10–44% of pulmonary adenocarcinomas.^{1–3} First-line treatment options for patients with *EGFR*-mutation-positive non-small-cell lung cancer (NSCLC) include the *EGFR* tyrosine kinase inhibitors (TKIs) gefitinib, erlotinib, and afatinib. Regulatory approval of these drugs was based on randomised phase 3 trials comparing each *EGFR* TKI with platinum-based chemotherapy.^{4–9} In addition to an improvement in response and progression-free survival, patients generally experience less toxicity and improvement in quality of life with *EGFR* TKIs than with platinum-based chemotherapy.^{10,11}

Unlike the first-generation *EGFR* TKIs (gefitinib and erlotinib), which are reversible inhibitors that selectively target *EGFR*, the second-generation *EGFR* TKIs afatinib and dacomitinib (PF-00299804) are irreversible inhibitors that have activity against all three kinase-active members of the ErbB family (*EGFR*/HER1, HER2, and HER4).^{12–15} The advantages of second-generation *EGFR* TKIs as first-line therapy for patients with *EGFR*-mutation-positive NSCLC have been clearly shown against chemotherapy but, when studied head-to-head against a first-generation TKI, the benefits were unclear.^{16,17}

In the single-arm phase 2 ARCHER 1017 study of dacomitinib as first-line therapy in patients with advanced NSCLC, 75.6% (95% CI 60.5–87.1) of patients with

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Research in context

Evidence before this study

Before planning the ARCHER 1050 study, we were aware that in the single-arm phase 2 ARCHER 1017 study of dacomitinib as first-line treatment in patients with advanced non-small-cell lung cancer (NSCLC), the subgroup of patients with an *EGFR*-activating mutation (exon 19 deletion or Leu858Arg) had a median progression-free survival of 18.2 months and 75.6% had an objective response, whereas patients with wild-type *EGFR* or unknown *EGFR* mutation status had a shorter progression-free survival and fewer objective responses. We searched PubMed using the search string “(*EGFR* AND lung cancer) AND (phase 2 or phase II or phase 3 or phase III)”, and limited our search to reports published between Jan 1, 2012, and April 19, 2017. We reviewed publications identified for head-to-head clinical trials on *EGFR* tyrosine kinase inhibitors (TKIs) evaluated in first-line advanced *EGFR*-mutation-positive NSCLC. The only relevant publications we found were on the phase 2b LUX-Lung 7 trial, which compared afatinib with gefitinib, and a phase 3 randomised controlled study of erlotinib versus gefitinib in Chinese patients with *EGFR*-mutation-positive advanced NSCLC (NCT01024413).

Added value of this study

To our knowledge, ARCHER 1050 is the first randomised phase 3 study to directly compare a second-generation *EGFR* TKI with a standard first-generation *EGFR* TKI for first-line treatment of patients with advanced *EGFR*-mutation-positive NSCLC. The primary endpoint, progression-free survival as assessed by masked independent review, was significantly longer with dacomitinib than with gefitinib. The duration of response in responders, by independent review, was longer with dacomitinib than with gefitinib. The safety profile of dacomitinib was consistent with previous studies. The incidences of diarrhoea and skin-related adverse events were higher with dacomitinib than with gefitinib.

Implications of all the available evidence

Patients with newly diagnosed advanced *EGFR*-mutation-positive NSCLC treated with first-line dacomitinib had significant improvements in progression-free survival and duration of response compared with gefitinib. Dacomitinib should be considered as a new first-line treatment option in patients with *EGFR*-mutation-positive NSCLC.

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EGFR-activating mutations achieved an objective response, with a median progression-free survival of 18.2 months (95% CI 12.8–23.8).¹⁸ Here we report the results of the randomised, open-label, phase 3 ARCHER 1050 study, which evaluated the efficacy and safety of dacomitinib versus gefitinib as first-line therapy in patients with advanced *EGFR*-mutation-positive NSCLC.

Methods

Study design and participants

ARCHER 1050 was an international, multicentre, randomised, open-label, phase 3 trial done at 71 universities and academic medical centres in seven countries or special administrative regions (China, Hong Kong, Japan, South Korea, Poland, Italy, and Spain; appendix pp 2–3). Eligible patients were aged at least 18 years old (or ≥ 20 years in Japan and South Korea), with histologically or cytopathologically confirmed newly diagnosed stage IIIB/IV or recurrent NSCLC (minimum of 12 months disease-free interval between completion of adjuvant or neoadjuvant therapy and recurrence of NSCLC), with at least one target lesion that had not previously been irradiated and was measurable according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 criteria, and with the presence of at least one documented *EGFR* mutation (exon 19 deletion or the Leu858Arg mutation, with or without the Thr790Met mutation). Patients were tested for their *EGFR* mutation status before randomisation. *EGFR* mutation status in tumour specimens from patients enrolled outside China was ascertained at local laboratories and was retested and

confirmed with the US Food and Drug Administration (FDA)-approved Qiagen therascreen *EGFR* RGQ PCR kit (version 2; Manchester, UK) at central laboratories. *EGFR* mutation status in tumour specimens from patients enrolled within China was determined at a central laboratory using version 1 of the Qiagen therascreen *EGFR* RGQ PCR kit and later retested and confirmed with the FDA-approved Qiagen therascreen *EGFR* RGQ PCR kit (version 2) at central laboratories in China. Other inclusion criteria included an Eastern Cooperative Oncology Group (ECOG) performance status of 0–1; adequate renal, hepatic, and haematological function; and availability of tumour specimens for central laboratory confirmation of an *EGFR* activating mutation. Patients were ineligible if they had any evidence of mixed histology, cytology, or both that included elements of small cell or carcinoid lung cancer; atypical *EGFR* mutations; history of brain or leptomeningeal metastases; history of, or currently suspected, diffuse non-infectious pneumonitis or interstitial lung disease; any previous anticancer systemic treatment of locally advanced or metastatic NSCLC; previous treatment with an *EGFR* TKI or other TKI; or uncontrolled or substantial cardiovascular disease.

The institutional review board or ethics committee of each participating institution approved the trial protocol. The trial was conducted in accordance with the International Conference on Harmonisation Good Clinical Practice guidelines and the provisions of the Declaration of Helsinki. All patients provided written informed consent before enrolment.

See Online for appendix

Randomisation and masking

Eligible patients were randomly assigned 1:1 to receive dacomitinib or gefitinib. Randomisation was stratified by race (self-reported; Japanese *vs* Chinese *vs* other east Asian *vs* non-Asian) and *EGFR* mutation type (exon 19 deletion *vs* Leu858Arg mutation). A randomisation list was generated using a computer-generated random code that was assigned by a central interactive web response system (IWRS). The IWRS was managed by a vendor (Cenduit Services; Bangalore, India) who had no clinical involvement with the trial. The allocation sequence, based on a randomisation-requirement specification form (prepared by the IWRS vendor in accordance with the requirements of the study sponsor), was generated by the IWRS. The investigators at the clinical sites enrolled the patients by using the IWRS, entered each patient's race and *EGFR* mutation type (the stratification variables), and assigned each patient to a treatment group on the basis of the IWRS output. In this open-label study, investigators and patients were not masked to treatment assignment. Tumour assessment by independent review was masked. Central imaging was masked to the reviewers by BioClinica (Princeton, NJ, USA). Sites were unmasked to study drug but the vendors and sponsor remained masked. Study sponsor personnel were unmasked at database lock for the primary analysis of progression-free survival.

Procedures

Patients in the dacomitinib group received oral dacomitinib 45 mg once daily in 28-day cycles. Dacomitinib dose reductions for a maximum of two dose levels were permitted for treatment-related toxicity in the case of grade 3 or worse toxicity, or prolonged grade 2 adverse events lasting more than one cycle. The first dose reduction was to 30 mg/day and the second dose reduction to 15 mg/day. After dose reduction, based upon investigator assessment, treatment was resumed at the dose level per protocol. Dose interruptions (<2 weeks, or longer in consultation with the sponsor) were permitted per protocol.

Patients in the gefitinib group received oral gefitinib 250 mg once daily in 28-day cycles. Gefitinib was only available as a 250 mg dose. If treatment was interrupted for grade 3, grade 4, or intolerable grade 2 toxicity, gefitinib was resumed at a daily or every-other-day dosing at the investigator's discretion.

In both study groups, treatment was discontinued after progression of disease, initiation of new anticancer therapy, unacceptable toxicities, non-compliance, withdrawal of consent, or death. Treatment beyond radiological progression was permitted as long as there was evidence of clinical benefit as determined by the investigator in consultation with the sponsor.

Tumour imaging assessments (CT or MRI) were done at screening, at the end of cycles one and two, and then at every other cycle until the end-of-treatment visit.

Objective tumour responses were measured using RECIST version 1.1 and assessed by a masked independent radiological central (IRC) review and by the investigator. Patients who discontinued treatment in the absence of progression were expected to be followed up every 8 weeks until disease progression. Haematology, blood chemistry, concomitant medications, and adverse events were assessed on day 1 of each 28-day cycle. Adverse events (including laboratory abnormalities) were assessed using the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) version 4.0, and left ventricular ejection fraction was also assessed as a safety outcome. Patient-reported outcomes were assessed at days 1 (baseline), 8, and 15 of cycle one, on day 1 of subsequent cycles, at the end-of-treatment visit, and at the post-treatment follow-up visit, using the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire-Core 30 items (EORTC QLQ-C30),^{19,20} the corresponding lung cancer module (QLQ-LC13),²¹ and the EuroQoL Group 5-Dimension Self-Report Questionnaire (EQ-5D).²² Further details on the patient-reported outcome assessments are in the appendix (p 4).

Outcomes

The primary endpoint was progression-free survival as determined by masked IRC review (defined as the time from randomisation to the date of disease progression according to RECIST version 1.1 per masked IRC review or death due to any cause, whichever occurred first) in the intention-to-treat population (all randomised patients). Secondary efficacy endpoints included progression-free survival based on investigator assessment, the proportion of patients who achieved an objective response (defined as a best overall response of either complete response or partial response, where best overall response is the best response recorded from the start of treatment until disease progression) based on both masked IRC review and investigator assessment, duration of response (defined as the time from first documentation of objective response to the date of disease progression or to death due to any cause, whichever occurred first) based on both masked IRC review and investigator assessment, overall survival (defined as the time from randomisation to the date of death for any cause), and overall survival at 30 months (defined as the probability of a patient being alive at 30 months from date of randomisation).

Additional secondary endpoints were safety assessment and patient-reported outcomes. Patient-reported outcome endpoints included overall change from baseline (longitudinal analysis) and time to deterioration in pain, dyspnoea, fatigue, or cough on the QLQ-C30 and QLQ-LC13 scales.

We also assessed time to treatment failure (defined as the time from randomisation to the date of treatment failure [first documentation of progression, death due to

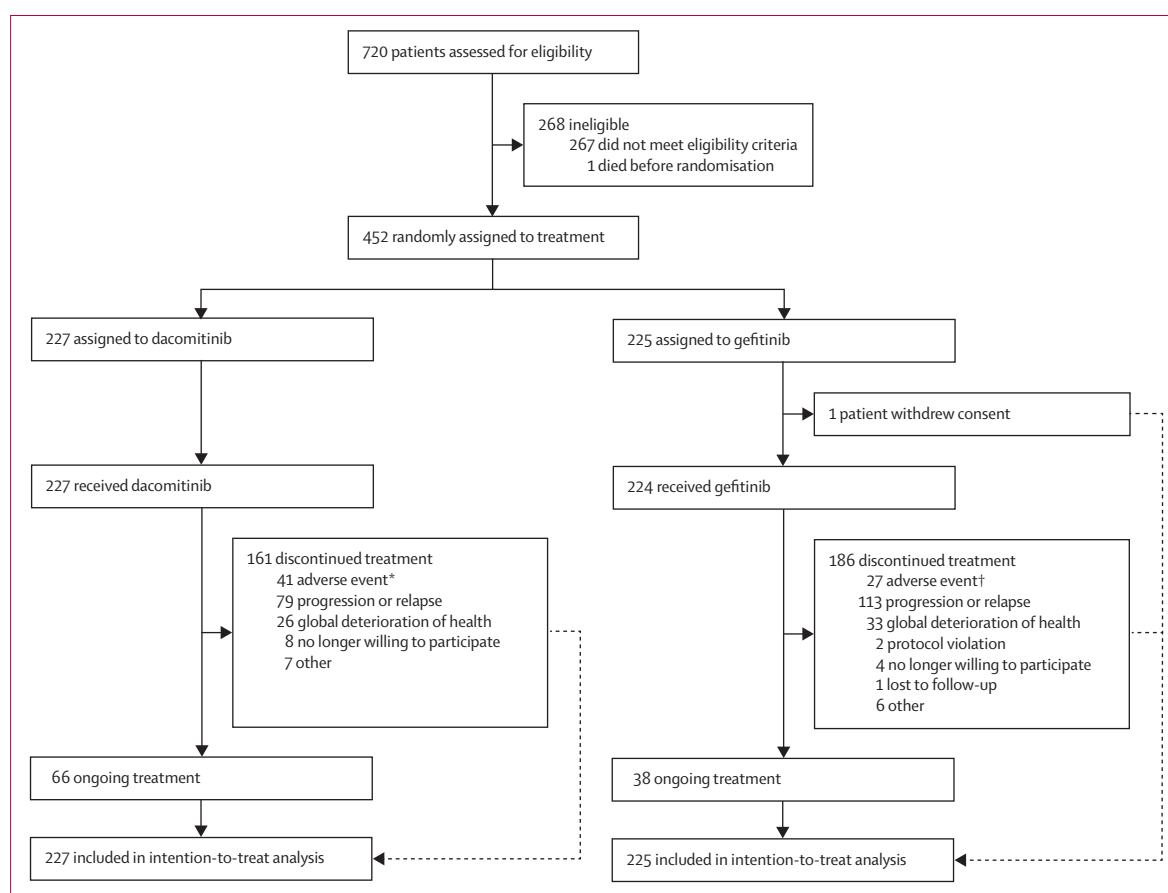


Figure 1: Trial profile

*22 treatment-emergent adverse events related to study drug, 18 treatment-emergent adverse events not related to study drug, and one non-treatment-emergent adverse event. †15 treatment-emergent adverse events related to study drug and 12 treatment-emergent adverse events not related to study drug.

any cause, or discontinuation of treatment due to any cause, whichever occurred first)) based on both masked IRC review and investigator assessment and restricted mean survival time (defined as the expected value of the variable $\min[T, \tau]$, where T is the time to event variable under consideration and truncation time τ is the timepoint of interest. Restricted mean survival time can be interpreted as the expected time of a patient being event-free out of the follow-up time of τ) based on both masked IRC review and investigator assessment. Time to treatment failure and restricted mean survival time were not specified in the protocol but added as secondary endpoints to the statistical analysis plan, version 3, on April 6, 2016.

Statistical analysis

We estimated that about 440 randomised patients with a minimum of 256 progression-free survival events observed would be required to achieve a 90% power to detect a 50% or more improvement in progression-free survival in patients in the dacomitinib group versus those in the gefitinib group in the intention-to-treat population (ie, hazard ratio [HR] ≤ 0.667),^{18,23} using a stratified log-rank test at a one-sided significance level of 0.025. Per the

statistical analysis plan, final overall survival analysis will occur after a minimum of 201 deaths.

All randomised patients (the intention-to-treat population) were included in the efficacy analysis. A log-rank test, stratified by *EGFR* mutation status at randomisation and race, was used to assess progression-free survival, time to treatment failure, and duration of response. A Cox proportional hazards model, stratified by *EGFR* mutation status and race as used in the log-rank test, was used to calculate HRs and 95% CIs for progression-free survival and time to treatment failure in the intention-to-treat population and duration of response among the objective responders in the intention-to-treat population. p values were determined by the log-rank test with adjustment for the same stratification factors. All reported p values were two-sided. We used the Schoenfeld residuals test to test the proportional hazards assumption.²⁴ Prespecified subgroups for subgroup analyses were age (<65 years vs ≥ 65 years), sex, ECOG performance status (0 vs 1), race (Asian vs non-Asian), smoking history (never vs former or current), and *EGFR* mutation type at randomisation (exon 19 deletion vs Leu858Arg). We also did a post-hoc exploratory analysis

	Dacomitinib (n=227)	Gefitinib (n=225)
Age, years		
Median	62 (53–68)	61 (54–68)
<65	133 (59%)	140 (62%)
≥65	94 (41%)	85 (38%)
Sex		
Male	81 (36%)	100 (44%)
Female	146 (64%)	125 (56%)
Race (self-identified)		
White	56 (25%)	49 (22%)
Black	1 (<1%)	0
Asian	170 (75%)	176 (78%)
Japanese	40 (18%)	41 (18%)
Chinese	114 (50%)	117 (52%)
Other east Asian	16 (7%)	18 (8%)
ECOG performance status		
0	75 (33%)	62 (28%)
1	152 (67%)	163 (72%)
Disease stage at screening		
Stage IIIB	18 (8%)	16 (7%)
Stage IV	184 (81%)	183 (81%)
Unknown*	25 (11%)	26 (12%)
Smoking status		
Never	147 (65%)	144 (64%)
Former	65 (29%)	62 (28%)
Current	15 (7%)	19 (8%)
Type of EGFR mutation†		
Exon 19 deletion‡	134 (59%)	133 (59%)
Leu858Arg§	93 (41%)	92 (41%)

Data are median (IQR) or n (%). ECOG=Eastern Cooperative Oncology Group.
 *Newly diagnosed with stage IV at time of study entry. †EGFR mutations (at randomisation) were identified from tumour specimens. ‡At randomisation, two patients in the gefitinib group (and none in the dacomitinib group) had the Thr790Met mutation. §At randomisation, two patients in the dacomitinib group (and none in the gefitinib group) had the Thr790Met mutation.

Table 1: Baseline characteristics (intention-to-treat population)

of progression-free survival in patients who responded to treatment within the Asian subgroup and non-Asian subgroup. HRs and p values for progression-free survival in subgroups were estimated from the unstratified Cox regression model and unstratified log-rank test, respectively. The proportions of patients achieving objective responses were compared between groups using Pearson's χ^2 test.

Patients in the intention-to-treat population who received at least one dose of study drug were included in the safety analysis. Adverse events were summarised using the Medical Dictionary for Regulatory Activities (MedDRA; version 19.1) preferred terms. An independent data and safety monitoring committee oversaw the trial and assessed patient safety on a periodic basis (after the first 60 patients were treated and every 6 months thereafter until study end).

For patient-reported outcomes, we used repeated-measures mixed-effects modelling to compare the two treatment groups with respect to the overall change from baseline scores on the QLQ-C30 and QLQ-LC13 scales using two-sided tests that were not adjusted for multiple testing. Patient-reported outcomes were evaluated in patients in the intention-to-treat population who also had a baseline assessment and at least one post-baseline assessment. Time to deterioration was defined as the time from randomisation to the first time the patient's score showed a 10 point or greater increase after baseline (indicating worsening condition) in any of the patient-reported symptoms of pain (chest; arm or shoulder), dyspnoea, fatigue, or cough on the QLQ-C30 and QLQ-LC13 scales or censored at date of the last questionnaire. A score increase of 10 points or more must have been held for at least two consecutive cycles for the symptom to be considered as deteriorated. We used the Kaplan-Meier method to estimate the time to deterioration and compared between treatment groups using the Hochberg-adjusted log-rank test. We did a sensitivity analysis without the condition of two consecutive cycles using the same methods and summary statistics.

Statistical analyses were done with SAS version 9.4. This study is registered with ClinicalTrials.gov, number NCT01774721.

Role of the funding source

ARCHER 1050 was designed by a steering committee of academic advisers and representatives from the sponsors of the study, SFJ Pharmaceuticals, and Pfizer (appendix p 3). Data were collected by the principal investigators (Y-LW, YC, XZ, KHL, KN, SNI, JC, MRM, AP, and TSM) and verified by the sponsors of the study. The sponsors had a role in the data analysis, data interpretation, and writing of the report. All the authors, including those employed by the sponsors of the study, vouch for the completeness and accuracy of the data and the data analyses and adherence to the protocol. All the authors had access to the data, prepared the report, and reviewed and approved the final submission. The corresponding author had final responsibility for the decision to submit for publication.

Results

Between May 9, 2013, and March 20, 2015, of 720 patients screened at enrolment, 452 were eligible and randomly assigned to receive either dacomitinib (227 patients) or gefitinib (225 patients; figure 1). One patient randomly assigned to the gefitinib group received no treatment because of rapid disease progression and consent was withdrawn; therefore, the safety analysis population comprised 451 patients. Demographic characteristics and baseline clinical characteristics were generally well balanced between the treatment groups (table 1).

Median duration of treatment at data cutoff (July 29, 2016) was 15.3 months (IQR 6.9–20.9) in the

dacomitinib group and 12.0 months (7.3–18.4) in the gefitinib group. At data cutoff, 66 (29%) patients in the dacomitinib group and 38 (17%) in the gefitinib group were still receiving study treatment.

Median duration of follow-up for progression-free survival in the intention-to-treat population, at data cutoff, was 22.1 months (95% CI 20.3–23.9; dacomitinib group 22.1 months [20.2–23.9] and gefitinib group 23.0 months [20.3–25.8]). According to masked IRC review, 136 (60%) of 227 patients in the dacomitinib group and 179 (80%) of 225 patients in the gefitinib group had progressed. Median progression-free survival was 14.7 months (95% CI 11.1–16.6) in the dacomitinib group and 9.2 months (9.1–11.0) in the gefitinib group (HR 0.59 [95% CI 0.47–0.74]; $p<0.0001$; figure 2A). We noted no evidence of non-proportional hazards based on the Kaplan-Meier curves and the Schoenfeld residuals plot of progression-free survival per IRC review (appendix p 12). Estimated progression-free survival per IRC review at 24 months was 30.6% (95% CI 23.8–37.5) in the dacomitinib group and 9.6% (5.6–15.0) in the gefitinib group. Progression-free survival based on investigator assessment was consistent with progression-free survival according to IRC review (figure 2B).

Subgroup analyses of progression-free survival based on IRC review were conducted in patient subgroups according to prespecified baseline characteristics; the findings were generally consistent with our main analysis (figure 3). We also did a post-hoc exploratory analysis of progression-free survival in Asian and non-Asian patients who responded to treatment (259 [75%] of 346 Asian patients and 72 [68%] of 106 non-Asian patients achieved a best overall response of complete or partial response; appendix p 13). In a prespecified subgroup analysis, we assessed progression-free survival, by IRC review, in patients with the exon 19 deletion at randomisation and in patients with the Leu858Arg mutation at randomisation (appendix p 14).

The proportion of patients who achieved an objective response according to masked IRC review was similar between the two groups (table 2, figure 4). The number of patients with the exon 19 deletion achieving an objective response, based on IRC review, was 102 of 134 patients (76%; 95% CI 68–83) in the dacomitinib group and 93 of 133 patients (70%; 61–78) in the gefitinib group ($p=0.2541$). For patients with the Leu858Arg mutation, these numbers were 68 of 93 patients (73%; 95% CI 63–82) in the dacomitinib group and 68 of 92 patients (74%; 64–83) in the gefitinib group ($p=0.9025$). In patients who responded to treatment, the duration of response based on IRC review was longer in the dacomitinib group than in the gefitinib group (HR 0.40, 95% CI 0.31–0.53, $p<0.0001$; table 2). Results based on investigator assessment were consistent with those based on IRC review (171 of 227 patients [75%; 95% CI 69–81] in the dacomitinib group had an objective response vs

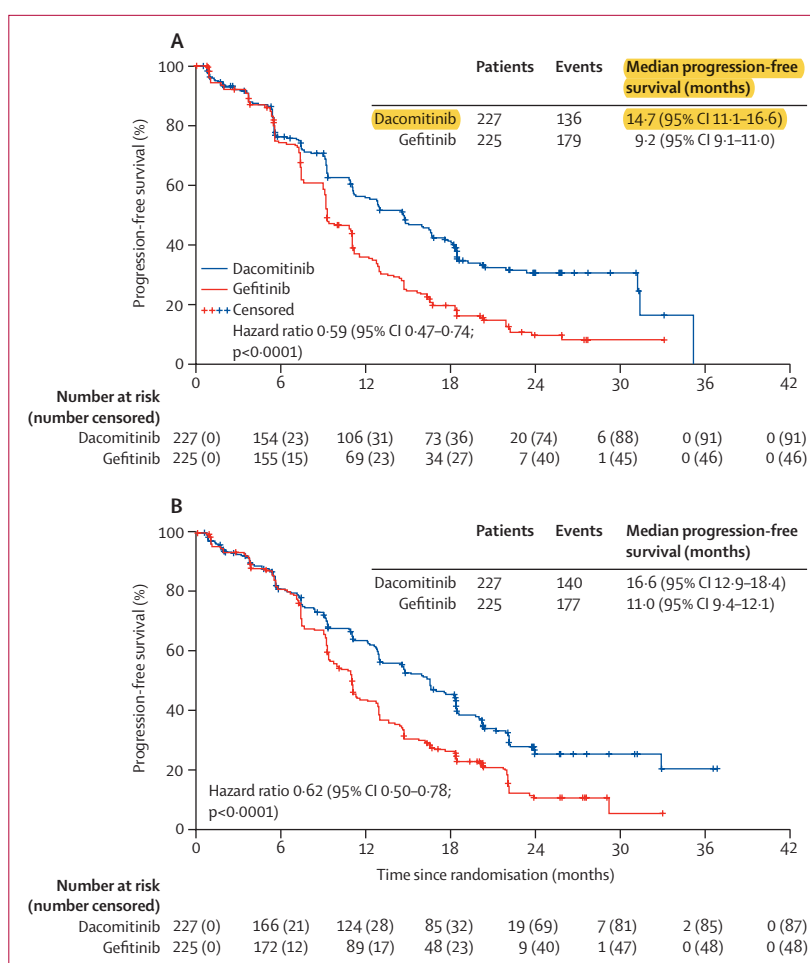


Figure 2: Progression-free survival (intention-to-treat population)
Assessed by masked independent review (A) and investigators (B).

158 of 225 patients [70%, 64–76] in the gefitinib group, $p=0.2224$; median duration of response was 15.9 months [95% CI 13.8–17.6] in the dacomitinib group vs 9.2 months [8.2–11.0] in the gefitinib group, HR 0.55, 95% CI 0.42–0.71, $p<0.0001$).

Analysis of time to treatment failure based on IRC review showed that patients in the dacomitinib group remained longer on study treatment than did those in the gefitinib group (median time to treatment failure 11.1 months [95% CI 9.2–14.6] in the dacomitinib group vs 9.2 months [7.6–9.4] in the gefitinib group; HR 0.67, 95% CI 0.54–0.83, $p=0.0001$). Results based on investigator assessment were consistent with those according to IRC review (median 13.0 months [95% CI 11.1–16.6] vs 11.0 months [9.3–11.1]; HR 0.70, 95% CI 0.56–0.86, $p=0.0006$). Restricted mean survival time based on IRC review with a truncation time of 33.1 months (selected as the minimum of the maximum progression-free survival time in the two groups) was longer with dacomitinib than with gefitinib (16.9 months [95% CI 15.3–18.5] vs 11.9 months [10.7–13.1]; difference

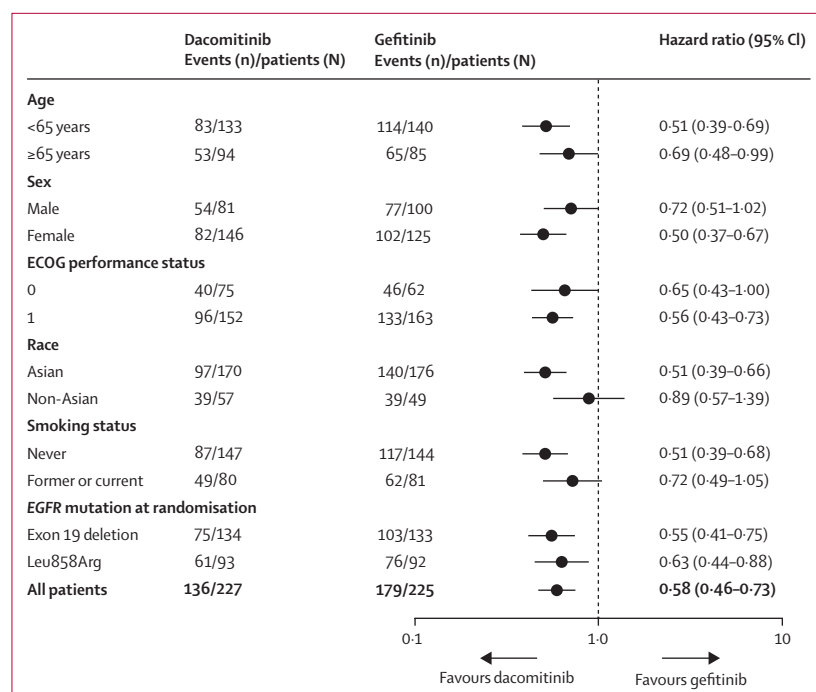


Figure 3: Subgroup analysis of progression-free survival (intention-to-treat population)
Assessed by masked independent review. ECOG=Eastern Cooperative Oncology Group.

	Dacomitinib (n=227)	Gefitinib (n=225)	p value
Type of response			
Complete response	12 (5%)	4 (2%)	..
Partial response	158 (70%)	157 (70%)	..
Stable disease	30 (13%)	27 (12%)	..
Progressive disease	12 (5%)	15 (7%)	..
Could not be evaluated	15 (7%)	22 (10%)	..
Objective response	170 (75%; 69–80)	161 (72%; 65–77)	0.4234*
Median duration of response, months†	14.8 (12.0–17.4)	8.3 (7.4–9.2)	<0.0001‡

Data are n (%; 95% CI) or median (95% CI). *Two-sided Pearson's χ^2 test. †Calculated with the Kaplan-Meier method from time of the first documented response until date of progression or the last Response Evaluation Criteria in Solid Tumors (RECIST) assessment for patients who did not have disease progression. ‡Two-sided Cochran-Mantel-Haenszel test, stratified by EGFR mutation status at randomisation (exon 19 deletion vs Leu858Arg mutation) and by race (Japanese vs Chinese and other east Asian vs non-Asian).

Table 2: Best overall response by masked independent review (intention-to-treat population)

5.0 [95% CI 3.0–7.0], $p<0.0001$). Restricted mean survival time based on investigator assessment was consistent with these results (17.4 months [15.9–19.0] vs 13.1 months [12.0–14.4]); difference 4.3 [95% CI 2.3–6.2], $p<0.0001$).

At data cutoff, overall survival data were not mature, with only 167 events (deaths) having occurred (76 [33%] of 227 patients in the dacomitinib group and 91 [40%] of 225 patients in the gefitinib group).

Post-progression systemic treatment was received by 93 (41%) of 227 patients in the dacomitinib group and 126 (56%) of 224 patients in the gefitinib group. The most common post-progression systemic treatments in

both study groups were pemetrexed, carboplatin, cisplatin, and osimertinib (appendix p 5).

Adverse events of any cause occurred in 226 (>99%) of 227 patients in the dacomitinib group and in 220 (98%) of 224 patients in the gefitinib group (table 3, appendix pp 6–8). The most commonly reported adverse events (of any grade) in patients given dacomitinib were diarrhoea (198 [87%]), paronychia (140 [62%]), dermatitis acneiform (111 [49%]), and stomatitis (99 [44%]). The most commonly reported adverse events (of any grade) in patients given gefitinib were diarrhoea (125 [56%]), alanine aminotransferase increase (88 [39%]), and aspartate aminotransferase increase (81 [36%]). The most commonly reported grade 3–4 adverse events were dermatitis acneiform (31 [14%] of 227 patients given dacomitinib vs none of 224 patients given gefitinib), diarrhoea (19 [8%] vs two [1%]), and raised alanine aminotransferase levels (two [1%] vs 19 [8%]). The frequency of grade 4 adverse events of any cause was similar in both groups (five [2%] in each group; table 3). Serious adverse events of any cause were reported in 62 (27%) of 227 patients given dacomitinib and in 50 (22%) of 224 patients given gefitinib (appendix p 9). Treatment-related serious adverse events were reported in 21 (9%) patients given dacomitinib and in ten (4%) patients given gefitinib. Treatment-related serious adverse events were gastrointestinal disorders (ten [4%] patients given dacomitinib vs none given gefitinib), skin and subcutaneous disorders (three [1%] vs none), respiratory disorders (two [1%] vs three [1%]), hepatobiliary disorders (three [1%] vs two [1%]), elevated liver enzymes (none vs three [1%]), metabolism and nutrition disorders (two [1%] vs none), infections (two [1%] vs one [$<1\%$]), keratitis (one [$<1\%$] vs none), acute kidney injury (one [$<1\%$] vs none), and chronic myeloid leukaemia (one [$<1\%$] vs none).

Deaths recorded by the investigators as adverse events occurred in 22 (10%) patients in the dacomitinib group and were disease progression (eight), pneumonia (two), respiratory failure (two), and one each of broncho-pulmonary aspergillosis, cerebral infarction, death (recorded as such in the case report form), diarrhoea, lung infection, metastases to meninges, multiple organ dysfunction syndrome, overdose due to diazepam, pneumonitis, and urinary tract infection. In the gefitinib group, deaths recorded by the investigators as adverse events occurred in 20 (9%) patients and were disease progression (11), dyspnoea (two), and one each of cerebral infarction, death (recorded as such in the case report form), general physical health deterioration, malignant neoplasm progression, malnutrition, pleural effusion, and pneumonia. During the treatment period and up to 28 days after the last dose, three patients died from study treatment toxicity: two in the dacomitinib group (one related to untreated diarrhoea and one related to untreated cholelithases/liver disease) and one in the gefitinib group (related to sigmoid colon diverticulitis/rupture complicated by pneumonia).

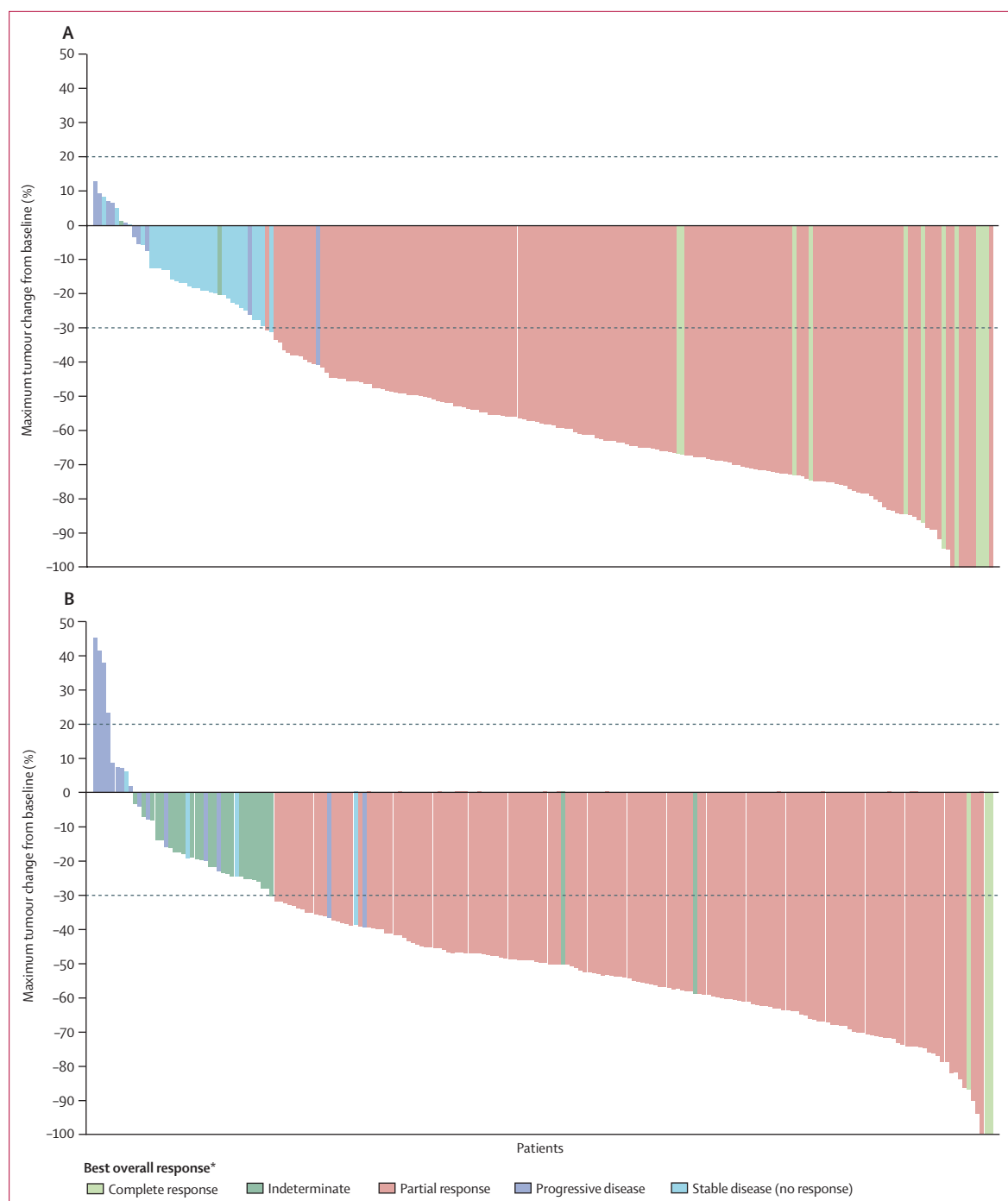


Figure 4: Maximum tumour change from baseline by best overall response (intention-to-treat population)

(A) Dacomitinib. (B) Gefitinib. Each bar represents an individual patient's maximum reduction in target lesion size, dashed lines show cutoffs for progressive disease ($\geq 20\%$ increase) and partial response ($\geq 30\%$ reduction). *Per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1, by masked independent radiological central review. Indeterminate was defined as progression not documented within 12 weeks after start of treatment date and where none of the other categories (complete response, partial response, stable disease, or progressive disease) was applicable.

Permanent discontinuation because of adverse events related to study drug occurred in 22 (10%) of 227 patients in the dacomitinib group and in 15 (7%) of 224 patients in the gefitinib group. The most frequent of these in the

dacomitinib group were skin and subcutaneous tissue disorders (n=7), gastrointestinal disorders (n=4), and interstitial lung disease or pneumonitis (n=3), whereas in the gefitinib group they were alanine aminotransferase or

	Dacomitinib (n=227)				Gefitinib (n=224)			
	Grades 1–2	Grade 3	Grade 4	Grade 5	Grades 1–2	Grade 3	Grade 4	Grade 5
Any adverse event	83 (37%)	116 (51%)	5 (2%)	22 (10%)	128 (57%)	67 (30%)	5 (2%)	20 (9%)
Diarrhoea	178 (78%)	19 (8%)	0	1 (<1%)	123 (55%)	2 (1%)	0	0
Paronychia	123 (54%)	17 (7%)	0	0	42 (19%)	3 (1%)	0	0
Dermatitis acneiform	80 (35%)	31 (14%)	0	0	64 (29%)	0	0	0
Stomatitis	91 (40%)	8 (4%)	0	0	39 (17%)	1 (<1%)	0	0
Decreased appetite	63 (28%)	7 (3%)	0	0	54 (24%)	1 (<1%)	0	0
Dry skin	60 (26%)	3 (1%)	0	0	38 (17%)	0	0	0
Weight decreased	53 (23%)	5 (2%)	0	0	36 (16%)	1 (<1%)	0	0
Alopecia	52 (23%)	1 (<1%)	0	0	28 (13%)	0	0	0
Cough	48 (21%)	0	0	0	41 (18%)	1 (<1%)	0	0
Pruritus	44 (19%)	1 (<1%)	0	0	28 (13%)	3 (1%)	0	0
ALT increased	42 (19%)	2 (1%)	0	0	69 (31%)	19 (8%)	0	0
Conjunctivitis	43 (19%)	0	0	0	9 (4%)	0	0	0
Nausea	40 (18%)	3 (1%)	0	0	48 (21%)	1 (<1%)	0	0
AST increased	42 (19%)	0	0	0	72 (32%)	9 (4%)	0	0
Rash	30 (13%)	10 (4%)	0	0	24 (11%)	0	0	0
Palmar-plantar erythrodysesthesia syndrome	31 (14%)	2 (1%)	0	0	7 (3%)	0	0	0
Pain in extremity	31 (14%)	0	0	0	26 (12%)	0	0	0
Dyspnoea	25 (11%)	4 (2%)	1 (<1%)	0	24 (11%)	4 (2%)	0	2 (1%)
Asthenia	24 (11%)	5 (2%)	0	0	25 (11%)	3 (1%)	0	0
Constipation	29 (13%)	0	0	0	31 (14%)	0	0	0
Mouth ulceration	28 (12%)	0	0	0	13 (6%)	0	0	0
Maculopapular rash	18 (8%)	10 (4%)	0	0	26 (12%)	1 (<1%)	0	0
Upper respiratory tract infection	25 (11%)	3 (1%)	0	0	28 (13%)	0	0	0
Musculoskeletal pain	24 (11%)	2 (1%)	0	0	28 (13%)	0	0	0
Dermatitis	21 (9%)	4 (2%)	0	0	8 (4%)	1 (<1%)	0	0
Insomnia	23 (10%)	1 (<1%)	0	0	33 (15%)	0	0	0
Anaemia	20 (9%)	2 (1%)	0	0	11 (5%)	5 (2%)	0	0
Chest pain	22 (10%)	0	0	0	32 (14%)	0	0	0
Hypokalaemia	11 (5%)	9 (4%)	2 (1%)	0	9 (4%)	4 (2%)	0	0
Vomiting	18 (8%)	2 (1%)	0	0	29 (13%)	0	0	0
Back pain	18 (8%)	0	0	0	34 (15%)	1 (<1%)	0	0
Pustular rash	6 (3%)	8 (4%)	0	0	3 (1%)	0	0	0
Hypertension	10 (4%)	3 (1%)	0	0	6 (3%)	4 (2%)	0	0
Disease progression	0	0	0	8 (4%)	0	0	0	11 (5%)
Pleural effusion	1 (<1%)	5 (2%)	0	0	4 (2%)	1 (<1%)	0	1 (<1%)
Lymphocyte count decreased	0	5 (2%)	0	0	2 (1%)	0	0	0
Abnormal hepatic function	2 (1%)	0	0	0	3 (1%)	4 (2%)	0	0

Data are n (%). ALT=alanine aminotransferase. AST=aspartate aminotransferase. The table lists all-cause, maximum grade adverse events reported in at least 10% of patients in either treatment group at grades 1–2, and adverse events reported at grade 3–5 in at least 2% of patients in either treatment group. Appendix p 6 shows all-cause, maximum grade adverse events reported in at least 10% of patients in either treatment group at grades 1–2 and all adverse events reported at grade 3–5.

Table 3: Adverse events (safety population)

aspartate aminotransferase increase (n=6) and interstitial lung disease or pneumonitis (n=3). Temporary discontinuations occurred in 177 (78%) patients in the dacomitinib group (median duration of temporary discontinuation 0·5 months [IQR 0·2–0·9]) and 120 (54%) patients in the gefitinib group (median duration of temporary discontinuation 0·2 months [0·1–0·5]).

Dose reductions occurred in 150 (66%) of 227 patients in the dacomitinib group; 87 (38%) patients received a

lowest dose of 30 mg/day, and 63 (28%) patients received a lowest dose of 15 mg/day. Patients were permitted to increase the dose after tolerating the lower dose and six patients had dose re-escalation. In the dacomitinib group, the median duration of dose reduction was 11·3 months (IQR 4·8–18·9) and the median time to the first dose reduction was 2·8 months (1·3–4·2). In the gefitinib group, dose reductions (every other day dosing) occurred in 18 (8%) of 224 patients, with a median

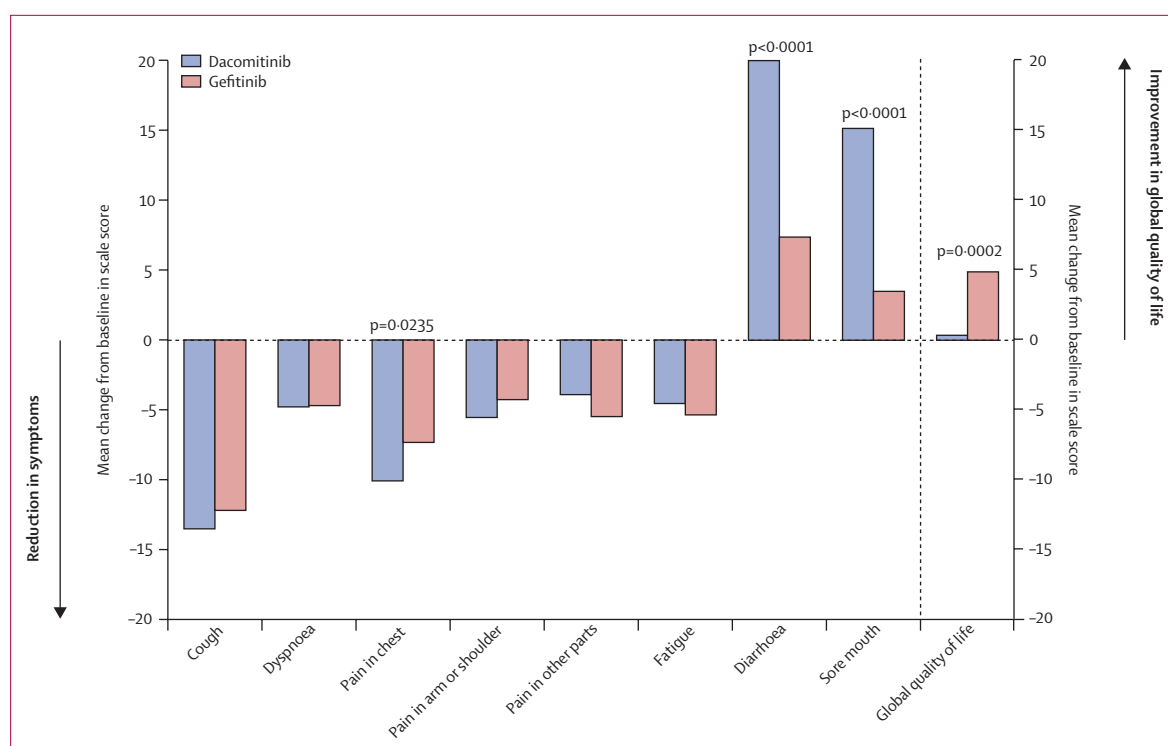


Figure 5: Overall change from baseline in key lung cancer-associated symptoms, fatigue, diarrhoea, sore mouth, and global quality of life

Each scale ranges from 0 to 100, with changes ≥ 10 points regarded as clinically meaningful. For global quality of life, higher scores indicate better global quality of life; for symptoms, higher scores indicate greater severity of symptoms. p values (unadjusted for multiple testing) are for the between-group comparison of the overall change from baseline, calculated using repeated-measures mixed-effects modelling.

duration of dose reduction of 5.2 months (IQR 2.5–7.9) and a median time to the first dose reduction of 3.3 months (2.4–4.2).

Patient-reported outcomes on baseline scores for the QLQ-C30, QLQ-LC13, and EQ-5D questionnaires are summarised in the appendix (p 10). The rates of completion were high, with more than 90% of patients answering all questions for almost all cycles in both treatment groups (data not shown). In the mixed model for repeated-measures analysis, we noted an overall improvement from baseline in the lung cancer symptom of pain in chest in both groups, which was greater in the dacomitinib group (mean -10.24 for dacomitinib vs -7.44 for gefitinib; $p=0.0235$; figure 5). Similar improvements from baseline in both groups were seen in lung cancer symptoms of dyspnoea (-4.89 vs -4.81 ; $p=0.9411$), cough (-13.61 vs -12.28 ; $p=0.3440$), pain in arm or shoulder (-5.58 vs -4.34 ; $p=0.2854$), and pain in other parts (-4.05 vs -5.49 ; $p=0.3288$; figure 5). Dacomitinib was associated with a greater increase from baseline in treatment-related symptoms of diarrhoea (19.88 vs 7.32 ; $p<0.0001$) and sore mouth (15.09 vs 3.51 ; $p<0.0001$; figure 5).

The overall change from baseline within each treatment group reached the 10-point threshold for being clinically meaningful²⁵ for decreased pain in chest, and increased diarrhoea and sore mouth in dacomitinib-treated

patients, and decreased cough in both dacomitinib-treated and gefitinib-treated patients. Although a statistically significant difference in global quality of life was observed between the two treatment groups, favouring gefitinib (0.20 for dacomitinib vs 4.94 for gefitinib; $p=0.0002$), it was maintained in the dacomitinib group and showed a statistically significant but not clinically meaningful improvement in the gefitinib group (figure 5). There was no difference in time to deterioration between the dacomitinib group and the gefitinib group with respect to pain (chest; arm or shoulder), dyspnoea, fatigue, or cough as a composite endpoint, nor its individual symptoms items (appendix pp 11, 15–16).

Discussion

To our knowledge, ARCHER 1050 is the first randomised phase 3 study to directly compare a second-generation EGFR TKI with a standard first-generation EGFR TKI for the first-line treatment of patients with advanced EGFR-mutation-positive NSCLC. Patients in the dacomitinib group had a significantly longer progression-free survival than did those in the gefitinib group. The median progression-free survival in the dacomitinib group is among the highest recorded in all registered randomised phase 3 studies with EGFR TKIs for EGFR-mutation-positive NSCLC.^{4–9,26} ARCHER 1050 supports the efficacy of dacomitinib as first-line therapy for

patients with *EGFR*-mutation-positive NSCLC and offers a new treatment option in this setting.

The results of the LUX-Lung 7 study,¹⁶ which compared afatinib with gefitinib as first-line treatment of patients with *EGFR*-mutation-positive NSCLC, have been previously reported. Unlike LUX-Lung 7, which had no specific statistical power for the three co-primary endpoints, ARCHER 1050 was prospectively powered to show a difference on a single primary endpoint. The improvement in progression-free survival, based on IRC review, with dacomitinib versus gefitinib (HR 0.59; median progression-free survival 14.7 months [95% CI 11.1–16.6] for dacomitinib vs 9.2 months [9.1–11.0] for gefitinib) in ARCHER 1050 was greater than that with afatinib versus gefitinib (HR 0.73; 11.0 months [10.6–12.9] for afatinib vs 10.9 months [9.1–11.5] for gefitinib) in LUX-Lung 7. The estimated progression-free survival at 24 months with dacomitinib (30.6%) in ARCHER 1050 was higher than that with afatinib (17.6%) in LUX-Lung 7.

In ARCHER 1050, dacomitinib's effect in prolonging the progression-free survival in the intention-to-treat population and in improving the duration of response in patients who responded to treatment was not accompanied by an increase in the proportion of patients achieving an objective response. The longer progression-free survival with dacomitinib than with gefitinib could be explained by the greater reduction in tumour size with dacomitinib treatment than with gefitinib treatment. The longer duration of response recorded with dacomitinib than with gefitinib in patients who responded to treatment might be due to the irreversible binding of dacomitinib to its targets in contrast to the reversible binding of gefitinib.

In our subgroup analysis in ARCHER 1050, the benefit of dacomitinib treatment compared with gefitinib in the non-Asian subgroup seemed to be lower than the benefit in the Asian subgroup of patients, although a post-hoc exploratory analysis suggested that dacomitinib was favoured over gefitinib in the 68% of non-Asian patients who responded (complete or partial) to treatment (appendix). The smaller sample size of the non-Asian subset (n=106) with a wide 95% CI versus that of the Asian subset (n=346) may have had an effect on the results. Additionally, in a single-arm phase 2 study,¹⁸ a subset analysis of patients similar to the population in the ARCHER 1050 study (patients with *EGFR*-activating mutations treated in a first-line setting) showed that the benefit in the Asian subset (median progression-free survival 16.2 months [95% CI 12.4–22.1]) seemed to be lower than that in the non-Asian subset (20.1 months [9.7–35.1]). Therefore, collectively the data do not suggest that non-Asian patients respond to dacomitinib treatment differently than do Asian patients.

Previous reports including a head-to-head phase 3 randomised controlled study of erlotinib versus gefitinib in Chinese patients with *EGFR*-mutation-positive

advanced NSCLC²⁷ have shown that *EGFR* TKIs have greater efficacy in patients with the exon 19 deletion than they do in patients with the Leu858Arg mutation.^{27–29} In the ARCHER 1050 study, the improvement in progression-free survival with dacomitinib compared with gefitinib was similar in the two subgroups, and the proportion of patients achieving an objective response with dacomitinib was similar in both subgroups (76% in patients with the exon 19 deletion and 73% in those with the Leu858Arg mutation). Our results are in agreement with the LUX-Lung 7 study, which showed no significant differences in efficacy (progression-free survival and proportion of patients with an objective response) with afatinib between the two mutation subgroups.¹⁶

In our analysis of time to treatment failure, we noted that time to treatment failure was shorter than progression-free survival in both treatment groups; this is probably because the possible types of events for progression-free survival (progression or death due to any cause) are a subset of the possible types of events for time to treatment failure (progression, death due to any cause, or discontinuation of treatment due to any cause).

The incidence of adverse events reported for dacomitinib in the ARCHER 1050 study is comparable with that reported for other dacomitinib studies,^{18,30–32} with no new safety signals identified. The incidence of adverse events was higher in the dacomitinib group than in the gefitinib group. Dose reductions occurred in 66% of patients in the dacomitinib group (compared with 8% in the gefitinib group). The effect of dacomitinib dose modification on efficacy is under evaluation as part of ARCHER 1050. We recommend that the dose of dacomitinib be adjusted according to the toxicity level. Prophylactic treatment with doxycycline and alclometasone, as demonstrated in another study of dacomitinib,³² might be beneficial. Prophylactic use of antidiarrhoeals such as loperamide may be helpful in managing diarrhoea associated with dacomitinib, followed by either dose interruptions or dose reductions.

In patient-reported measures of key disease-associated symptoms, the dacomitinib group showed a significantly greater overall improvement of pain in chest compared with the gefitinib group, while improvements in dyspnoea, cough, pain in arm or shoulder, and pain in other parts were similar in the two groups. As expected with respect to the toxicity profile, a significantly higher deterioration in the treatment-associated symptoms of diarrhoea and sore mouth was recorded in the dacomitinib group. Despite clinically meaningful deteriorations in diarrhoea and sore mouth in dacomitinib-treated patients, global quality of life was maintained with dacomitinib, and the difference in global quality of life, although statistically significant in favour of gefitinib, was small.

The ARCHER 1050 study has some limitations. First, the study had an open-label design, which could have

introduced bias. However, the results for progression-free survival, objective responses, and duration of response by investigator assessment were consistent with those based on independent review. Second, patients with brain metastases were excluded from participation and this criterion might have enriched our population for patients with better overall prognosis. The reason for this exclusion criterion was that the capacity of dacomitinib in brain penetration was not known at the time of the study design and there is a lack of adequate CNS penetration with gefitinib. Third, the dacomitinib group had a higher proportion of female patients and a higher proportion of patients with an ECOG performance status of 0 than the gefitinib group. These artifacts of randomisation were not considered limitations to the study or the results because sex is not a prognostic factor of progression-free survival in patients with NSCLC with *EGFR*-activating mutations and generally there is no difference on prediction of outcome between ECOG performance status of 0 and 1. Fourth, the subgroup analyses were not adjusted for multiple testing and were not adequately powered by study design.

In conclusion, dacomitinib treatment was superior to gefitinib with respect to progression-free survival and duration of response in the first-line treatment of patients with *EGFR*-mutation-positive NSCLC and should be considered as a new treatment option for this population.

Contributors

Y-LW, KN, FT, RL, RR, EIS, TW, JLW, and TSM contributed towards the study conception and design. Y-LW, YC, XZ, KHL, KN, SNI, JC, MRM, AP, and TSM contributed towards patient recruitment. SNa was involved in the data analysis and interpretation and manuscript writing. RS was involved in the evaluation of the patient-reported outcomes. All authors contributed towards data analysis and interpretation. All authors drafted and reviewed the report and approved the final version for submission.

Declaration of interests

Y-LW has received honoraria from AstraZeneca, Roche, and Eli Lilly. KN has received honoraria from Astellas, AstraZeneca, EPS Holdings, Ono, Kyowa Hakko Kirin, Showa Yakuhin Kako, SymBio, Daiichi Sankyo, Chugai, Boehringer Ingelheim, Eli Lilly, Pfizer, Bristol-Myers Squibb (BMS), Novartis, Kissei, Taiho, Merck Sharp & Dohme (MSD), and Ayumi, and research funding from Chugai, Ono, EPS Associates, Quintiles, Daiichi Sankyo, Eisai, PPD-SNBL, Pfizer, Takeda, Boehringer Ingelheim, Taiho, BMS, GlaxoSmithKline, AstraZeneca, Kyowa Hakko Kirin, AbbVie, Novartis, Eli Lilly, Yakult Honsha, PAREXEL, Otsuka, Astellas, AC Medical, and Merck Serono. SNI has received honoraria from Eli Lilly, AstraZeneca, Taiho, BMS, Boehringer Ingelheim, Chugai, and Yakult Honsha, and research funding from Eli Lilly, AstraZeneca, Novartis, MSD, and Chugai. FT is an employee of SFJ Pharmaceuticals, Japan. RL is an employee of SFJ Pharmaceuticals, USA. MRM has received honoraria from AstraZeneca, Boehringer Ingelheim, and BMS, and consulting fees from AstraZeneca, Boehringer Ingelheim, and BMS. AP has received consulting fees from Boehringer Ingelheim; fees for serving on advisory boards from AstraZeneca, Roche, Boehringer Ingelheim, MSD, and BMS; and research funding from Pfizer and BMS. EIS, TW, JLW, SNa, and RS are employees of and own stock in Pfizer. TSM has received honoraria from AstraZeneca, Roche/Genentech, Eli Lilly, BMS, Boehringer Ingelheim, Novartis, MSD, and Pfizer; fees for serving on advisory boards from AstraZeneca, Roche/Genentech, Eli Lilly, Merck Serono, BMS, Pfizer, Boehringer Ingelheim, Novartis, Clovis Oncology, Vertex, SFJ Pharmaceuticals, ACEA BioSciences, MSD, geneDecode, Oncogenex, Celgene, and Ignyta; research funding from AstraZeneca, Boehringer Ingelheim, Pfizer,

Novartis, SFJ, Roche/Genentech, MSD, Clovis Oncology, BMS, Taiho, and Eisai; and owns stock in Sanomics. YC, XZ, KHL, RR, and JC declare no competing interests.

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